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1. Project Introduction

The **Sales Analytics Dashboard** is a Power BI-based business intelligence project developed to analyse sales performance, profitability, regional performance, product categories, sales representatives, and discount impact.

The dashboard provides management with a consolidated view of key performance indicators such as **Total Sales, Total Profit, Profit Margin, and Total Orders**. It also enables users to analyse monthly sales trends, regional performance, category-wise sales, sales representative performance, and the relationship between discounts and profit.

The objective of this project is to convert raw retail sales data into meaningful business insights that can support data-driven decision-making.

2. Problem Statement

ABC Retail Pvt Ltd. operates across multiple regions and sells products belonging to different categories and subcategories.

Management was facing difficulty in understanding:

- Which regions are performing well?
- Which categories generate the highest sales?
- Which regions generate the highest profit?
- How are sales changing over time?
- Which sales representatives are performing better?
- How profitable are different product categories?
- Whether discounts are affecting profitability?

Since the information was stored in raw transactional data, management needed an interactive dashboard to monitor performance and identify business opportunities.

3. Business Objectives

The main objectives of this project are:

1. Analyse overall sales performance.
2. Monitor total sales and total profit.
3. Calculate and analyse profit margin.
4. Identify high-performing and low-performing regions.

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5. Analyse monthly sales trends.
6. Compare category-wise sales performance.
7. Analyse category-wise profit margins.
8. Evaluate sales representative performance.
9. Analyse the relationship between discounts and profit.
10. Provide actionable recommendations to improve sales and profitability.
11. Create an interactive Power BI dashboard for management decision-making.

4. Dataset Information

The project uses retail and e-commerce transactional data along with supporting dimension tables.

Table	Purpose
Retail & Ecommerce	Main transaction/fact table used for sales, cost, profit, discount, region, and order-level analysis.
Category	Category master table used to organize products into reporting categories.
Subcategory	Subcategory master table used for more detailed product-level analysis.
Sales Representative	Sales representative master table used to evaluate individual sales performance.
Calendar Table	Date/timetable used for monthly, quarterly, and yearly trend analysis.

Main Retail Dataset Columns

- Order Id
- Order Date
- Month
- Quarter
- Year
- Region
- State
- Category
- Subcategory
- Sales
- Cost
- Profit
- Discount
- Sales Representative
- Remarks

5. Data Dictionary

Column	Description	Data Type
Order Id	Unique identifier for an order.	Text/Number
Order Date	Date on which the order was placed.	Date
Month	Month of the order.	Text
Quarter	Quarter of the order.	Text
Year	Year of the order.	Whole Number
Region	Geographical region of the transaction.	Text
State	State where the transaction occurred.	Text
Category	Product category.	Text
Subcategory	Product subcategory.	Text
Sales	Revenue generated from the order.	Decimal
Cost	Cost associated with the order.	Decimal
Profit	Profit generated from the order.	Decimal
Discount	Discount applied to the order.	Decimal
Sales Representative	Person responsible for the sale.	Text
Remarks	Additional transaction comments.	Text

Dimension Tables

Category Table

- Category Id
- Category

Subcategory Table

- Subcategory Id
- Subcategory

Sales Representative Table

- Sales Representative Id
- Sales Representative

6. Data Understanding

Column	Issue	Count	Action Required
Order ID	Null values	13 rows	Remove rows containing null values.
Order Date	Invalid data type: categorical instead of date	187 rows	Convert to date format. If conversion is not possible, remove the affected rows.
Order Date	Null values	91 rows	Replace missing values if a reliable source is available; otherwise, remove the affected rows.
Month	Inconsistent values	1,813 rows	Standardize month names by trimming spaces and applying proper case.
Quarter	No issues found	N/A	No action required.
Year	No issues found	N/A	No action required.
Region	Incorrect formatting and inconsistent values	2,056 rows	Correct inconsistent values by trimming spaces, standardizing names, and capitalizing each word.
State	Misspelled state names: Telengana, Maharastra, Karnatka	Telangana – 34 rows; Maharastra – 31 rows; Karnatka – 24 rows	Replace misspelled state names with the correct values.
State	Inconsistent formatting	1,927 rows	Trim spaces and capitalize each word.
State	Null values	129 rows	Replace null values with “Unknown” if no reliable source is available.
Category	Misspelled category names: Electronic, Furnitures, Officesupplies	Electronic – 180 rows; Furnitures – 136 rows; Officesupplies – 164 rows	Replace incorrect category names with the correct values.

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Column	Issue	Count	Action Required
Category	Inconsistent formatting	2,056 rows	Trim spaces and capitalize each word.
Subcategory	Null values	111 rows	Replace null values with “Unknown” if no reliable source is available.
Sales	Invalid values during decimal conversion	280 rows	Convert values to decimal. Remove rows that contain non-numeric values if they cannot be corrected.
Cost	Invalid values during decimal conversion	285 rows	Convert values to decimal. Remove rows that contain non-numeric values if they cannot be corrected.
Profit	Invalid values during decimal conversion	227 rows	Convert values to decimal. Remove rows that contain non-numeric values if they cannot be corrected.
Discount	No issues found	N/A	No action required.
Sales Representatives	Inconsistent names: R IYER, S.khan	R IYER – 121 rows; S.khan – 121 rows	Standardize names using the correct naming convention.
Sales Representatives	Null values	258 rows	Replace null values with “Unknown” if no reliable source is available.
Remarks	Inconsistent formatting	1,071 rows	Trim spaces and convert text to proper case.
Remarks	Null values	220 rows	Replace null values with “Unknown” if no reliable source is available.

7. Data Cleaning

Step	Transformation	Reason	Completed
Removed Blank Rows	Removed rows containing only blank values.	Blank rows do not contain useful information and may affect analysis.	Yes
Order Date	Changed the data type from text to date and removed null or invalid date values.	Date format is required for time-based reporting and analysis.	Yes
Month	Corrected inconsistent month names, trimmed spaces, and applied proper case.	Standardized month values improve consistency and reporting accuracy.	Yes
Region	Corrected inconsistent region names, trimmed spaces, and applied proper case.	Standardized region values support accurate regional comparison.	Yes
State	Corrected misspelled state names, trimmed spaces, applied proper case, and replaced missing values with "Unknown".	Clean state values improve location-based analysis and reduce reporting errors.	Yes
Category	Corrected incorrect category names, trimmed spaces, and applied proper case.	Standardized category names make category-wise analysis reliable.	Yes
Subcategory	Replaced null values with "Unknown".	Missing subcategory values were unavailable from the source data.	Yes
Sales	Changed the data type from text to decimal number and removed conversion errors.	Numeric data type is required for sales calculations and analysis.	Yes
Cost	Changed the data type from text to decimal number and removed conversion errors.	Numeric data type is required for cost calculations.	Yes
Profit	Changed the data type from text to decimal number and removed conversion errors.	Numeric data type is required for profit calculations.	Yes

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Step	Transformation	Reason	Completed
Sales Representative	Corrected inconsistent names, trimmed spaces, applied proper case, and replaced missing values with "Unknown".	Standardized representative names improve performance tracking.	Yes
Remarks	Replaced null values with "Unknown", trimmed spaces, and applied proper case.	Clean remarks improve readability and consistency across records.	Yes
Close & Apply	Applied all transformations and loaded the cleaned dataset into Power BI.	Finalized data preparation for analysis and visualization.	Yes

8. Exploratory Data Analysis (EDA)

EDA was performed to understand patterns and trends in the sales data.

The analysis focused on:

Sales Performance

Total sales were analysed across:

- Month
- Region
- Category
- Sales Representative

Profitability

Profit was analysed to identify:

- High-profit regions
- Low-profit regions
- High-margin categories
- Low-margin categories

Regional Analysis

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Sales and profit were compared across:

- East
- West
- North
- South

Category Analysis

The dashboard compares:

- Electronics
- Furniture
- Office Supplies

Sales Representative Analysis

Sales representatives were ranked according to total sales performance.

Discount Analysis

The relationship between discount and profit was explored using a scatter chart.

Create a table.

9. Dashboard Development

The dashboard was developed using **Microsoft Power BI**.

Dashboard KPIs

Four major KPI cards were created:

KPI	Dashboard Value
Total Sales	15.42M
Total Profit	4.67M
Profit Margin	30.26%
Total Orders	1K

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DAX Measures

Measure Name	DAX Formula	Purpose
Total Sales	Total Sales = SUM('Retail & E-commerce'[Sales])	Calculates total revenue generated from sales.
Total Profit	Total Profit = SUM('Retail & E-commerce'[Profit])	Calculates total profit earned from transactions.
Total Orders	Total Orders = DISTINCTCOUNT('Retail & E-commerce'[Order Id])	Counts the number of unique customer orders.
Profit Margin %	Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)	Measures profitability as a percentage of total sales.

Visualizations Developed

Visual	Purpose
KPI Cards	Display key business metrics
Line Chart	Monthly sales trend
Bar Chart	Sales and profit by region
Bar Chart	Sales by category
Bar Chart	Profit margin by category
Column Chart	Sales representative performance
Scatter Chart	Discount versus profit
Slicers	Year, Quarter and Region filtering

Dashboard Filters

The dashboard provides interactive filters for:

- Year
- Quarter
- Region

These filters allow management to analyse the business from different perspectives.

10. Key Findings

Based on the dashboard shown:

1. Strong Overall Sales

The business generated approximately **15.42M in total sales**.

2. Healthy Profitability

Total profit is approximately **4.67M**, with an overall profit margin of **30.26%**.

3. Regional Performance Differences

The dashboard shows noticeable differences in sales performance across regions. The **West region appears to be the strongest region by sales** in the displayed regional comparison.

4. Electronics Leads Category Sales

Electronics appears to generate the highest sales among the displayed categories, followed closely by Furniture.

5. Category Margins Are Relatively Similar

Profit margins across the major categories appear to be relatively close to each other, suggesting that no single category has an extremely different margin profile in the displayed data.

6. Sales Representative Performance Varies

The Sales Representative chart shows clear differences in sales contribution among representatives.

7. Unknown Representative Records

An **Unknown** category appears in the Sales Representative analysis, indicating missing or unmatched representative information.

8. Monthly Sales Fluctuate

The monthly trend shows variation in sales throughout the period, with some months performing better than others.

11. Business Insights

The dashboard provides the following business insights:

1. Overall Performance: The business generated approximately 15.42M in sales and 4.67M in profit, with an approximately 30% profit margin.

2. Regional Performance: West is currently the strongest region in terms of both sales and profit, while other regions show comparatively lower performance.

3. Sales Trend: Monthly sales fluctuate over time, indicating that management should investigate seasonal or operational factors affecting demand.

4. Product Performance: Categories contribute differently to revenue, so management should evaluate category performance using both sales and profitability.

5. Sales-

Team Performance: Sales representatives contribute differently to overall performance, making representative-level analysis useful for identifying top performers and improvement opportunities.

7. Monthly sales fluctuate: Your monthly trend shows noticeable variation.

12. Recommendations

Based on the analysis, the following recommendations can be provided:

1. Strengthen Underperforming Regions

Identify the reasons behind lower regional sales and improve:

- Marketing
- Product availability
- Sales coverage
- Customer engagement

2. Focus on High-Selling Categories

Maintain sufficient inventory for high-performing categories such as Electronics.

3. Optimize Discount Strategy

Discounts should not be increased simply to increase sales. Management should evaluate the **profit impact of each discount level**.

4. Improve Sales Representative Performance

Identify top-performing representatives and understand their sales practices.

Provide additional training and support to lower-performing representatives.

5. Fix Missing Sales Representative Data

Investigate the **Unknown** sales representative records and map them to the correct representatives.

6. Monitor Monthly Trends

Management should monitor monthly sales fluctuations and plan inventory and marketing campaigns around high- and low-performing periods.

13. Conclusion

The **Sales Analytics Dashboard** successfully transforms retail transaction data into an interactive Power BI business intelligence solution.

The dashboard provides management with a clear view of:

- **15.42M Total Sales**
- **4.67M Total Profit**
- **30.26% Profit Margin**
- **Approximately 1K Orders**

The analysis highlights differences in regional performance, category sales, sales representative performance, monthly trends, and discount-profit relationships.

The project demonstrates how **data cleaning, data modelling, DAX measures, exploratory analysis, visualization, and business intelligence** can be combined to support better business decisions.

Overall, the dashboard enables management to move from **raw data → analysis → insights → business decisions**.